



VCS VERIFICATION REPORT

INNER MONGOLIA BAOTOU BAYIN WIND
POWER PROJECT

(VCS PROJECT ID: 355)



Document Prepared By

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Summary:

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions reported for the project activity “Inner Mongolia Baotou Bayin Wind Power Project” (VCS Project ID:355) for the monitoring period 01/01/2013-29/08/2018 within the 1st VCS crediting period 30/08/2008-29/08/2018, to review and determine the monitored reductions in GHG emissions that have occurred as a result of the project activity. These emission reductions are claimed as Verified Carbon Units (VCU) under the VCS Standard Version 4.2.

The verification was performed on the basis of VCS Standard Version 4.2 for the VCS projects, as well as criteria given to provide for consistent project operations, monitoring and reporting. The verification was conducted by means of document review, follow-up interviews and site inspections, and the resolution of outstanding issues. The verification team identified 3 CLs and no CAR or FAR was raised in this monitoring period.

In CTI’s opinion, the GHG emission reductions reported for the project in the monitoring report (version 02 dated 30/05/2022) are fairly stated. The GHG emission reductions of the period from 01/01/2013-29/08/2018 within the 1st VCS crediting period were calculated correctly on the basis of approved methodology ACM0002 “Grid-connected electricity generation from renewable sources”

(version 6.0), the monitoring plan contained in the registered VCS PD (i.e registered CDM PDD (version 04, dated 25/07/2008) for the 1st VCS crediting period 30/08/2008-29/08/2018.

CTI does not assume any responsibility towards the issuance and utilization of the VCU's hereby verified and certified. Request for issuance of VCU's shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

Hence, CTI is able to certify that the emission reductions from the "Inner Mongolia Baotou Bayin Wind Power Project" during this monitoring period from 01/01/2013-29/08/2018 amount to 2,109,463 tCO₂e.

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1 INTRODUCTION

Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd. has commissioned Shenzhen CTI International Certification Co., Ltd (CTI) to carry out the verification and certification of emission reductions reported for the “Inner Mongolia Baotou Bayin Wind Power Project” (the Project) for the monitoring period 01/01/2013-29/08/2018. This report contains the findings from the verification and includes a verification statement for the verified carbon units.

1.1 Objective

Verification is the periodic independent review and ex-post determination by an accredited verification body of the monitored reductions in GHG emissions that have occurred as a result of the registered VCS project activity during a defined verification period.

A verification statement is the written assurance by a verification body that, during a specific period in time, a project activity achieved the emission reductions as verified.

The objective of this verification was to verify and provide a verification statement of emission reductions reported for the “Inner Mongolia Baotou Bayin Wind Power Project” for the period 01/01/2013-29/08/2018.

1.2 Scope and Criteria

The scope of the verification is:

- To verify that actual monitoring systems and procedures are in compliance with the monitoring systems and procedures described in the monitoring plan;
- To evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement;
- To verify that reported GHG emissions data is sufficiently supported by evidence.

The criteria of the verification are:

- VCS Program Guide (version 4.1)/39/
- VCS Standard (version 4.2)/38/ and other relevant requirements defined by Verra;
- The approved methodology ACM0002 (version 6.0)/42/ applied by the project.

The verification shall ensure that reported emission reductions are complete and accurate in order to be verified.

1.3 Level of Assurance

The verification report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. As per the para 3.9.1 of the VCS Standard Version 4.2, the project size categorization belongs to 'Large Project' as the expected annual emission reductions is 500,678tCO_{2e} greater than 300,000 tCO_{2e}. Therefore, according to the requirement of para 4.1.8 (4) of VCS standard Version 4.2 "The threshold for materiality with respect to the aggregate of errors, omissions and misrepresentations relative to the total reported GHG emission reductions and/or removals shall be five percent for projects and one percent for large projects.", CTI applied a materiality threshold of 1% with respect to omission or misstatements concerning reported quantities of the project activity.

1.4 Summary Description of the Project

Sectoral Scope and Project Type

According to the VCS Program Guide (version 4.1)/39/ and Annex A of the Kyoto Protocol, the project is applicable under the following activity categories:

- Sectoral Scope: 1. Energy (Renewable/non-renewable)
- Project Type: wind power project

Project Background

The project is located in Damao United Banner (County), Baotou City, Inner Mongolia Autonomous Region, P.R.China. The site location's approximate coordinates are north latitude 42°20'12" and east longitude 109°16'32". The project activity was registered as a CDM project with Ref.2153 on 12/05/2009 and the renewable crediting period was selected starting on 12/05/2009. The project has been registered as VCS project with Ref. VCS 355 under VCS Standard Version 2007.1 with the crediting period start date 30/08/2008 when the commencement of operations activities started as confirmed by checking the Statement on the timeline of the project /32/, operation logs /7/ and the previous VCS monitoring and verification report.

The amount of 173,742 tCO_{2e} VCU's have been issued for the monitoring period 30/08/2008-11/05/2009 within the 1st VCS crediting period. GHG emission reductions have been applied for CERs and under CDM of which total 1,705,459 tCO_{2e} have been issued for the monitoring periods 12/05/2009-30/06/2009, 01/07/2009-31/12/2009, 01/01/2010-30/04/2010, 01/05/2010-30/11/2010, 01/12/2010-30/04/2011, 01/05/2011-30/04/2012 and 01/05/2012-31/12/2012.

The installed capacity of the project is 201MW equipped with 134 sets of wind generator in type of Gold wind 77/1500kW, with unit capacity of 1500kW, manufactured by Xinjiang Gold Wind Science and Technology Co., Ltd. The electricity generated by the project activity is supplied to the North China Power

Grid (hereafter referred to as “NCPG”), and the project is estimated to deliver 500,678 tCO₂e emission reductions annually.

2 VERIFICATION PROCESS

2.1 Method and Criteria

The verification was performed through means of the following three phases in accordance with the requirement of the registered VCS PD (i.e. registered CDM PDD version 04 dated 25/07/2008) and monitoring plan contained in the registered VCS PD, the applied methodology, and the VCS Standard (version 4.2) and other relevant VCS requirements:

- A desk review of the monitoring report and all support documents;
- Follow-up interviews with project stakeholders and site inspection;
- The resolution of outstanding issues and the issuance of the verification report and statement.

The following sections outline each step in more detail.

The verification of the emission reductions has assessed all factors and issues that constitute the basis for emission reductions from the project. These include:

- The emission reduction calculations and the relevant data records;
- The calibration and maintenance records for the monitoring instruments;
- The management systems to support the project operation and monitoring.

2.2 Document Review

Based on the requirements of competency, experience and qualified sectoral scopes, CTI appointed a verification team in accordance with CTI’s internal procedures.

Function	Name	Technical competence	Task Performance*
Team Leader& Verifier	Feng Tian	1.2	☒ DR ☒ SV ☒ RP ☐ TR
Technical Reviewer	Lin Shunrong	1.2, 14.1, 15.1	☒ DR ☐ SV ☐ RP ☒ TR

*DR=Document review; SV=Site visit; RP=Reporting; TR=Technical review

In addition to the VER/VCU monitoring report/1/, the registered VCS PD /11/, emission reduction calculation spreadsheet/2/ during this monitoring period, the following documents also were assessed as a part of the verification audit:

- The VCS validation report /23/;

- The previous VER/VCU monitoring report/25/ and verification report/26/, and previous CDM/CER monitoring reports /27/and verification reports/28/;
- Baseline and monitoring methodology ACM0002 (version 6.0) applied by the project/42/;
- Relevant decisions, clarifications and guidance from the Verra/38/-/41/; and
- Other information and references relevant to the project activity.

During the desk review, CTI has applied standard auditing techniques to assess the quality of information provided. The following activities were performed:

- A review of the data and information presented to verify their completeness;
- A review of the monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, the quality of metering equipment including calibration requirements, and the quality assurance and quality control procedures; and

An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions.

2.3 Interviews

On 18/05/2022, CTI visited the project proponent Longyuan (Baotou) Wind Power Co., Ltd. to perform on-site assessment. The key personnel of the project were interviewed or assisted the verification team /42/. Main topics of the interview cover implementation of the project construction, applicability of selected methodology, implementation of project monitoring, emission reduction calculation, etc.

The key personnel interviewed /43/ are summarized in the table below:

Interviewed personnel	Role	Organization	Subject
Mr. Zhang Bin	Plant Director	Longyuan (Baotou) Wind Power Co., Ltd.	Operation of the project activity; Implementation of the monitor plan of the project activity; Data collection and data achievement; Calibration of meters and equipment maintenance; The complaint by local stakeholders; The stakeholder consultation during the operation of the project activity. Project's sustainable development contributions
Mr. Li lei	O&M Staff		
Mr. Li Xiaolei	O&M Staff		
Mr. Chen Xijin	Local resident	Local Village	The impact of the project activity; The complaint by local stakeholders; The stakeholder consultation during the operation of the project activity; Project's sustainable development contributions.
Mr. Liu Jianmin			
Mr. Huang Puyi	Staff	Local Environment Protection Bureau	
Mr. Xue Qiang	Project Manager	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.	Data collection and ER calculation.

2.4 Site Inspections

During the on-site assessment, CTI has applied standard auditing techniques to assess the quality of information provided. The following aspects of the project activity have been verified:

- An assessment of the implementation and operation of the registered project activity is as per the registered VCS PD of the project activity;
- A review of information flows for generating, aggregating and reporting the monitoring parameters; and
- Interviews with relevant personnel to determine whether the operational and data collection procedures are implemented in accordance with the monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks and the electricity receipts;
- A check of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered VCS PD and the selected methodology;
- A review of calculations and assumptions made in determining the GHG data and emission reductions; and

- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

In addition, all parameters required by the monitoring methodology ACM0002 (version 6.0), and the management system were assessed during the site visit.

2.5 Resolution of Findings

A corrective action request (CAR) shall be raised, where:

- i. Non-conformities with the monitoring plan or methodology are found in monitoring and reporting, or if the evidence provided to prove conformity is insufficient;
- ii. Mistakes have been made in applying assumptions, data or calculations of emission reductions which will impair the estimate of emission reductions;
- iii. Issues identified in a FAR during validation to be verified during verification have not been resolved by the project proponents.

A clarification request (CL) shall be raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

The verification team identified 3 CLs and no CAR in this monitoring period. The CLs were satisfactorily addressed by the project proponents in the updated monitoring report version 02 dated 30/05/2022 (refer to Appendix A for further details).

2.5.1 Forward Action Requests

A forward action request (FAR) is issued for actions if the monitoring and reporting require attention and/or adjustment for the next monitoring period.

CTI confirmed that there was no FAR identified in previous verification/26/, and no FAR was raised during this verification.

2.6 Eligibility for Validation Activities

The project has been validated by DET NORSKE VERITAS under the CDM rules and registered as CDM project on 12/05/2009, and was registered as VCS project under VCS standard version 2007.1 through the gap validation by Bureau Veritas Certification Holding SAS.

CTI was accredited in Sectoral Scope 1 on validation and verification and thus is eligible for verification activities on the project. By checking the UNFCCC website, the VCS pipeline and the previous verification reports/28/, it is confirmed the verification for this monitoring period 01/01/2013-29/08/2018 which is no more than consecutive 6 years is the first verification undertaken by CTI. Therefore, the verification undertaken by CTI is in line with the requirement for VVB stipulated in Section 4.1.20 of VCS standard version 4.2

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

The project has been registered as a CDM project on 12/05/2009 under the UNFCCC with Ref. 2153. The renewable CDM crediting period was selected starting on 12/05/2009. By checking the UNFCCC website: <https://cdm.unfccc.int/Projects/DB/DNV-CUK1218624488.68/view>, it is confirmed that total 1,705,459tCO₂e of CERs have been issued for the monitoring periods 12/05/2009-30/06/2009, 01/07/2009-31/12/2009, 01/01/2010-30/04/2010, 01/05/2010-30/11/2010, 01/12/2010-30/04/2011, 01/05/2011-30/04/2012 and 01/05/2012-31/12/2012.

3.2 Methodology Deviations

The validation process /16/ has assessed all factors and issues that constitute the basis for emission reductions from the project according to the applicable CDM methodology ACM0002 (version 6.0) /42/. There was not any methodology deviation applied to this project. Details refer to section 4.1.

3.3 Project Description Deviations

By checking the previous VCS verification report with the monitoring period 30/08/2008-11/05/2009, and the Statement on the timeline of the Project /32/, and through the site visit interview with project owner, it is confirmed that the project started operation on 30/08/2008 which is the project start date according to the requirement of VCS. A deviation was identified for the crediting period of the Project. The project is registered as VCS project under VCS version 2007.1 and completed validation before 19/03/2020. As per VCS requirement, it remains eligible to apply the crediting period requirements under VCS Version 3 which shall be a maximum of ten years and may be renewed at most twice. Therefore, the first renewable crediting period of the Project under VCS is updated from 30/08/2008-11/05/2009 to 30/08/2008-29/08/2018, which lasts for 10 years.

As per para 3.8.9-3 of VCS Standard version 4.2, the project did not conduct the validation of crediting period renewal within the two years after the end of the first VCS crediting period. Thus, the crediting period of the project ends on 29/08/2018.

CTI confirms that the project remains in compliance with VCS Program rules and this deviation has no impact on the applicability of the methodology, additionality or the appropriateness of the baseline scenario and meet all appropriate rules and requirements of VCS standard.

Through the site visit interview with project owner, physically checking the power connection system as well as the Statement on the deactivation of the 10kV backup line issued by the grid company/31/, it is confirmed that the 10kV backup line has been deactivated on 25/12/2012 and no more electricity was imported through the backup line. Thus, the net electricity supplied to the grid by the project (EG_y) and

net electricity supplied to the grid through the main line (EG'y) are same, which is the difference between the electricity exports and imports through main line. Therefore, the parameter EG_{backupline} and EG'y have been removed from the parameters to be monitored.

The verification team confirmed that this deviation will not lower the accuracy and conservativeness of the monitoring system and will not negatively impact the conservativeness of the quantification of GHG emission reductions or removals. Furthermore, the deviation relates only to the criteria and procedures for monitoring or measurement, doesn't relate to any other part of the methodology.

CTI confirms that the project is implemented in accordance with the project description, and no any other project description deviations occurred. Details refer to section 4.1.

3.4 Grouped Project

The project was not a grouped project; hence this clause is not applicable.

4 VERIFICATION FINDINGS

This section summarizes the findings from the verification of the emission reductions reported for the "Inner Mongolia Baotou Bayin Wind Power Project" for the monitoring period 01/01/2013-29/08/2018.

4.1 Project Implementation Status

Project Implementation in accordance with the registered project design document

The project is a wind power grid-connected plant and locates Damao United Banner (County), Baotou City, Inner Mongolia Autonomous Region, P.R.China. The geographical coordinates are verified by GPS system as north latitude 42°20'12" and east longitude 109°16'32". The total installed capacity is 201MW equipped with 134 sets of wind generator in type of Gold wind 77/1500kW, with unit capacity of 1500kW. It was confirmed by checking the Statement on the timeline of the project /32/, operation logs/7/, the previous VCS monitoring report and verification report /25/-/26/ the first turbine was put into trail operation on 30/08/2008, and the project was put into full commercial operation on 13/04/2009. The actual implementation of the project during this verification period was verified in terms of nameplates of wind turbines /29/ and the equipment purchase contract /33/. The details of the turbines with respect to their installation and capacity have been verified to be consistent with description indicated in the registered VCS PD.

The electricity generated by the project activity is supplied to the NCPG, which can be confirmed by the Power purchase agreement (PPA) signed between Longyuan (Baotou) Wind Power Co., Ltd. and Inner Mongolia Electric Power Company/3/. All the monitoring system in operation period is consistent with the description in monitoring plan contained in the registered VCS PD. The control system at the power plant is automated and assures continuous operation, including monitoring on malfunction of equipment.

By checking the daily operation and maintenance records/7/ and via on-site visit, CTI can confirm that no serious malfunction happened and the plant was under a normal operation as expected in this monitoring period.

On-site training for the related procedures including, recording and reporting was verified to be in place/6/ and their implementation was confirmed by interview with the key operators/43/ and observing the operation.

CTI confirms that the project implementation is in accordance with the project description contained in registered VCS PD. Furthermore, through visual inspection and document review, it is confirmed that all physical features of the project activity including data collection systems and storage systems have been implemented in accordance with the monitoring plan.

CL 01 was raised to clarify the inconsistency on the starting date of operation for the first wind turbine with previous CDM verification which has been successfully closed. Details refer to Appendix D.

Compliance of monitoring plan with monitoring methodology

CTI is able to confirm that the monitoring plan contained in the VCS PD is in accordance with the approved methodology applied by the project activity, i.e. ACM0002 (version 6.0).

Compliance of monitoring with the monitoring plan

The monitoring has been carried out in accordance with the monitoring plan contained in the VCS PD/11/, through the site visit interview with project owner, physically checking the power connection system as well as the Statement on the deactivation of the 10kV backup line issued by grid company /31/, it is confirmed that the 10kV back up line has been deactivated on 25/12/2012 and no more electricity was imported through this 10kV backup line. Thus, EG_{backuptime} was not monitored during this monitoring period.

CTI confirms that all the other parameters stated in the monitoring plan are monitored and reported appropriately. Parameters required to be monitored by the monitoring plan as per the monitoring methodology ACM0002 (version 6.0) and the management system were assessed during the site visit. The monitoring report lists each parameter required by the monitoring plan and the information flow (i.e. from data generation, recording, aggregation, calculation and reporting) for these parameters is provided. The information flow for each parameter is further verified in the following sections.

Parameters fixed ex ante

“Data and parameters fixed ex ante” in the MR are checked against the registered VCS PD. The parameter EF_y is the combined emission factor of the grid, which was determined ex-ante at the validation stage and has been fixed during the first crediting period. CTI verified and confirms that the emission factor used in the monitoring report is in compliance with the VCS PD.

Parameters monitored

CTI conducted document review and performed on-site assessment with project stakeholders to:

- A review of information flows for generating, aggregating and reporting the monitoring parameters;
- Determine whether the operational and data collection procedures are implemented in accordance with the registered monitoring plan;
- A cross-check between information provided in the monitoring report and data from other sources such as plant logbooks, ETNs and Electricity confirmation letter;
- An identification that quality control and quality assurance procedures in place to prevent or identify and correct any errors or omissions in the reported monitoring parameters.

According to the monitoring plan and the actual monitoring undergoing during this monitoring period, the parameters to be monitored are as below:

EG_y Net electricity supplied by the project activity to the grid

Monitored parameter EG_y

EG_y is the net electricity supplied by the project to the grid through the main line which is calculated as the differences between the electricity exported to and imported from the grid through the main line. By site inspection and checking the monitoring plan contained in the registered VCS PD and previous verification reports, CTI confirmed electricity exported to and imported from the mainline are continuously measured by Meter 1 and its backup Meter 1' as well as Meter 2 and its backup Meter 2'. Meter 1 and Meter 1' are bi-directional and installed outlet of the transformer station (35kV/220kV) of the project which are controlled by PP. Meter 2 and Meter 2' are installed at the entrance to Wanghai 220 kV substation of the grid which are controlled by the grid company.

The verification team had on-site checked the location of the meters against the diagram of power connection system /4/ and found them to be consistent and installed in accordance with the approved monitoring plan. All the monitoring facilities and system have been verified by CTI during on-site visit.

By checking operation log /7/ and via on-site visit, it is confirmed that all meters were working well during this monitoring period. Via the on-site interview and by checking the daily operation and maintenance records/7/, monthly and daily reading records and ETNs and Electricity confirmation letter /9/ issued by grid company, it is confirmed by the verification team that the electricity is continuously measured by Meter 1 and its backup meter Meter1' , Meter2 and its backup meter Meter2'. Readings of Meter1 and Meter1' are daily recorded and monthly aggregated by PP while readings of Meter2 and Meter2' are daily recorded and monthly aggregated by the grid company. The cut-off time of electricity exports and imports is at 00:00 on the 21st day of each month. The grid company supplied the monthly aggregated readings from Meter2 to PP which were checked by PP with the readings of Meter1. After confirmed by both sides, the ETNs/9/ based on the readings of Meter 2 was issued by the grid company and have been used for cross-checking the records of electricity exports and imports. .

As described above, the meters have been installed in accordance with VCS Program rules and meet all appropriate rules and requirements of VCS standard. CTI has on-site checked the location of the meters against the diagram of power connection system and found them to be consistent.

Data in the monthly reading records were used to the report, through a cross check with ETNs or electricity confirmation letter. The most conservative values among readings of Meter1, Meter2 and values from ETNs were used for the monitoring report and ERs calculation spreadsheet, which has been verified by the verification team. Supporting references and data required to determine the quantity of net electricity generation supplied by the project is found to be complete and transparent.

Monitoring equipment and calibration

The documents review was carried out. Particular attention was paid to the frequency of measurements, the quality of the monitoring equipment including calibration performance and observations of monitoring practices against the requirements of the registered VCS PD and the methodology ACM0002(version 6.0) and corresponding tool(s), and the quality assurance and quality control procedures.

All monitoring meters have been calibrated annually as per the monitoring plan covering this monitoring period. The relevant information of the calibration is listed as below:

Item	Meter 1	Meter 1'	Meter 2	Meter 2'
Type	ZMQ202C.4r4af6	ZMQ202C.4r4af6	ZMQ202C.4r4af6	ZMQ202C.4r4af6
Accuracy	0.2S	0.2S	0.2S	0.2S
Serial No.	94471210	94471211	95151114	95151118
Calibration frequency	Annually			
Calibration date	10/11/2012 08/11/2013 06/11/2014 02/11/2015 30/10/2016 26/10/2017			
Calibration validity	09/11/2013 07/11/2014 05/11/2015 01/11/2016 29/10/2017			

	25/10/2018
Calibration entity	Inner Mongolia Electric Power Research Institute Electric Metrological Centre

Calibration records and accreditation certificates/12//13/have been verified by the verification team. The actual accuracy of all meters is verified to be 0.2S which is consistent with the requirement in the monitoring plan contained in the registered VCS PD and in compliance with the Technical administrative code of electric energy metering /15/. The calibration frequency of meters is annual, which is consistent with the monitoring plan. Based on the site visit and by checking the calibration reports, CTI confirms that:

- All meters have been installed in accordance with the monitoring plan;
- The meters' accuracy is in line with the requirement of the monitoring plan;
- The calibration frequency of the meters is annual, which is in line with the monitoring plan. And the calibrations are verified to be valid for the whole reporting period.

CL 02 was raised to clarify the type of the electricity meters which has been successfully closed. Refer to Appendix D for details.

Data management and control

Longyuan (Baotou) Wind Power Co., Ltd. is responsible for operation and routine maintenance of power plant under the project activity. The quality assurance and quality control procedures have been addressed in the project management and monitoring manual/5/, including the organization structure with the responsibilities, personnel competencies, monitoring procedures and monitoring management. By interview with the staff/43/ and check records /7//8/ during on-site visit, it can be confirmed that the monitoring management system is implemented following the project management and monitoring manual.

All monitoring devices have been calibrated and maintained periodically to ensure the accuracy of measurement. Calibration records of instruments used in measurements were made available during the verification visit and found to be valid for the entire period of the verification. Competence and training records of in-plant personnel engaged in measurement of plant parameters were presented during verification and found to be in order.

As the above and document review, verification team considers that the project activity operated properly in the way complying with the implementation plan and monitoring plan contained in the VCS PD and the monitoring was performed in a transparent and comprehensive manner during this monitoring period.

By checking the UNFCCC website: <https://cdm.unfccc.int/Projects/DB/DNV-CUK1218624488.68/view> , it is confirmed that emission reductions within the monitoring periods

12/05/2009-30/06/2009, 01/07/2009-31/12/2009, 01/01/2010-30/04/2010, 01/05/2010-30/11/2010, 01/12/2010-30/04/2011, 01/05/2011-30/04/2012 and 01/05/2012-31/12/2012 have been issued as CERs and no other issuance request was submitted under CDM.

Therefore, it is concluded no double counting issuance has occurred for this verified monitoring period 01/01/2013-29/08/2018 under both VCS and CDM programmes. Via the on-site interview and checking the Statement on no double counting made by PP /34/, it is confirmed that the emission reductions for this verified period are only requested for VCUs issuance.

The verification team has also checked other GHGs programs and forms of credit such as exchange platform of Chinese Certified Emission Reduction(CCER) projects, Gold Standard, EU ETS, IREC, China ETS and based on the interview with the project proponent, during this monitoring period, it is confirmed that except CDM and VCS scheme:

No evidence showing that the project has participated or been rejected under any other GHG programs since validation or previous verification

There is no observation that the project has received or sought any other emission trading program and form of environmental credit, or has become eligible to do so since validation or previous verification

The project is completely voluntary CDM and VCS project, not included in an emissions trading program or any other mechanism that includes GHG allowance trading. Although the national emission trading scheme has been launched since 16/07/2021, it only covers the high-emission and energy intensive industries that emit at least 26,000 tonnes of CO₂e/year as per the regulation/19/ including thermal power generation, petrochemical, chemical, building materials, iron and steel, etc. By checking the business license/14/of the project proponent, it is confirmed that the PP is qualified in business scope of renewable energy investment, thus is not included the mandatory emission control scheme and there is no emission cap enforced for the PP by further checking the compulsive company list under China ETS in public information/20/. Moreover, it is confirmed by CTI the project is not registered as a CCER project in China, the emission reductions generated by the project activity are not eligible for transaction under China ETS as per the regulation/19/. Hence, it is confirmed that the emission reductions will not be double counted.

Via the on-site inspection, and through the interviews with the project proponent, local villagers and official from local environment bureau, and by checking the evidences provided by the project proponent, the verification team agrees on the following points regarding to the promotion of sustainability development by the project activity:

- *SDG 7, Indicator 7.2.1 Renewable energy share in the total final energy consumption*

The project provides clean and renewable energy through making use of wind power, displacing fossil fuel power in the grid NCPG. During this monitoring period, 1,963,024.289 MWh of net electricity generated by wind energy was delivered to the NCPG. It is verified by checking the daily and monthly reading records/8/ and via on-site inspection. This contributes to increasing the share of clean energy consumption of China's "Action Plan for Carbon Dioxide Peaking Before 2030" /36/.

- *SDG 8, Indicator Numbers of decent work opportunities provided which has income higher than local/national average level*

It is verified by interviewing with project proponent, staff and local villagers and checking the Statement on the annual income of employees from 2013 to 2018 /21/ and Per capita disposable income of residents in Inner Mongolia Autonomous Region in 2013 and 2018 /22/, that the project activity provides 27 long-term job opportunities for local residents including 14 positions for women, and provides higher salary from 137,618 RMB/year in 2013 to 189,635 RMB/year in 2018 above the local average level of 25,497 RMB/year in 2013 and 28,376 RMB/year in 2018. Hence, the contribution to this SDG has been confirmed and the provisions for monitoring and reporting this goal is verified as reasonable. This also contributes to one of the China's actions for promoting sustainable developing, "Increase labor force participation rate through implementation of the classification policy. Vigorously enforce the Law on Promotion of Employment" /35/.

- *SDG 13, Indicator Tonnes of greenhouse gas emissions avoided or removed.*

As verified by on-site inspection, and via checking the emission reduction calculation sheet/2/ and the daily and monthly reading records/8/, 2,109,463 tCO₂e GHG emission reductions are generated by the project activity during this monitoring period. This contributes achieve one of China's stated sustainable development priorities "Actively adapt to climate change and strengthen resistance capacity to climate risks in agriculture, forestry, water resources and other key fields, as well as cities, coastal regions and ecologically vulnerable areas" /37/.

Meanwhile, by checking the approved VCS and CDM monitoring reports for previous monitoring periods, it is confirmed that the project contributions with all impacts were not included in previously approved VCS monitoring reports or reported in Sustainable Development Contribution Reports, thus, all impacts during the previous monitoring periods from 30/08/2008 to 31/12/2012 have not been calculated in the cumulative impact. Therefore, the cumulative contributions over the project lifetime are considered as the same as the current project contributions.

In conclusion, Verification Team was able to confirm that the project implementation is in accordance with the project description contained in the VCS PD. All required equipments and procedures are available and implemented in an appropriate manner. All necessary monitoring instruments are installed and operated. The project is completely operational and the same has been confirmed on-site. Neither mistakes nor malfunction on main meters have been observed during this monitoring period.

4.2 Safeguards

4.2.1 No Net Harm

Via on-site inspection and checking the EIA summary and conclusion provided in the approved VCS PD, it is confirmed that the impact caused by the project activity on the surrounding ecosystem and residents, wastewater, noise, solid waste and atmosphere etc. is very little, there would be no net harm caused due to the project activity. Appropriate mitigation measures have been undertaken by the project proponent to avoid the potential impacts during the construction and operation. The EIA of the project are approved by the government.

Also, no potential environment or social economic matter was found during the site visit. It is verified by interviewing with project proponent, staff and local villagers and checking the Statement on the annual income of employees from 2013 to 2018 /21/ and Per capita disposable income of residents in Inner Mongolia Autonomous Region in 2013 and 2018 /22/, that the project has contributed to the local employment and improved income levels by providing job opportunities and offering permanent employees wages well above local average income levels. As stated by the plant staff during the on-site interviews, they have gained knowledge and skills in the regular training organized by the project proponent, so it can be judged that the project has, to some extent, improved the marketability of the employees as well as the possibility of lifelong learning for them.

4.2.2 Local Stakeholder Consultation

The local stakeholder's meeting was held in Damao County on 10/07/2007. 28 participants attended the meeting including local residents, builders and members of the local authorities. The project owner introduced the proposed project, and then a survey was arranged through a questionnaire. The opinions expressed by the stakeholders were recorded and are available on request.

The stakeholder meeting and the survey showed that the proposed project receives strong support from the local community. They all believe the proposed project will promote local economic development and agree with the project development and construction.

For continuous communication with local stakeholders, the project owner carried questionnaire survey every two years to collect the relevant comments and suggestions, i.e. in June 2013, July 2015 and May 2017 during this monitoring period. For each survey carried out, 30 questionnaires have been handed and all of them have been received successfully from the stakeholder representatives. Via checking the filled questionnaires during on-site visit, it is confirmed by the verification team that no negative comments were received. Some comments shows that the local stakeholders have concerns about the pollution control and security of the power plant during operation. Through on-site visit and interviewing with staff from PP, it is confirmed that as a renewable energy project, the project merely causes environmental pollution and the pollution caused e.g waste, was properly handled. Furthermore, by checking the training records /6/ of the on-site staff, it is confirmed that all staff have received sufficient training before and during the implementation of the project so that the security of the power plant can be guaranteed. Through interviewing with staff from local environmental protection bureau and local residents during the site visit, it is confirmed by the verification team that during the implementation stage of this monitoring period, local authority has conducted spot checks on the implementation of the project periodically as per the request from the local governments' regulations, no negative comments and issues from the local stakeholders during this monitoring period were identified and the project passed all the periodic spot checks by local government.

All such conclusion has been verified through site visit and checking VCS PD.

4.3 AFOLU-Specific Safeguards

For non-AFOLU projects, this section is not required.

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

CTI confirms that appropriate methods and formulae for calculating baseline emissions, project emissions and leakage have been followed, and the assumptions, emission factors and default values that are applied in the calculation have been justified.

According to the applied methodology, the emission reductions are determined as the difference between the baseline emissions, project emissions and leakage:

$$ER_y = BE_y - PE_y - L_y$$

Baseline emissions

Baseline emissions are determined as multiplying quantity of net electricity supplied by the project to the grid (EG_y) by the validated ex-ante fixed grid emission factor (EF_y).

$$BE_y = EG_y \times EF_y$$

Grid emission factor (EF_y)

EF_y is the grid emission factor of the which has been verified ex-ante in the validation stage in the VCS PD as 1.07460 tCO₂e/MWh.

The net electricity supplied to the grid by the project (EG_y)

The verification team has checked the daily and monthly reading records, ETNs, Confirmation letter for electricity readings issued by the grid company and emission reduction calculation sheet, and confirmed the data and calculation applied are correct, complete and transparent. The reading records and EG_y calculation are summarized in Table 1 as follow:

Table 1 The net electricity supplied to the grid by the Project (EG_y)

Year	Month	Reading records of electricity exported to the grid via the main line from Meter1 (MWh)	Reading records of electricity exported to the grid via the main line from Meter2 (MWh)	Values of electricity exported to the grid via the main line from ETNs (MWh)	Conservative values of electricity exported to the grid via the main line (MWh)	Reading records of electricity imported from the grid via main line from Meter1 (MWh)	Reading records of electricity imported from the grid via main line from Meter2 (MWh)	Values of electricity imported from the grid via the main line from ETNs (MWh)	Conservative values of electricity imported from the grid via the main line (MWh)	EG _y (MWh)
-	-	A	B	C	D=min(A,B,C)	E	F	G	H=max(E,F,G)	I=D-H
2013	01-Jan~20-Jan	22,923.770	22,915.110	22,915.110	22,915.110	103.110	105.120	105.120	105.120	22,809.990
	21-Jan~20-Feb	29,442.642	29,424.430	29,424.430	29,424.430	116.510	117.740	117.740	117.740	29,306.690
	21-Feb~20-Mar	37,280.622	37,262.410	37,262.410	37,262.410	106.510	107.740	107.740	107.740	37,154.670
	21-Mar~20-Apr	57,146.872	57,128.660	57,128.660	57,128.660	144.460	145.690	145.690	145.690	56,982.970
	21-Apr~20-May	39,404.895	39,383.895	39,383.895	39,383.895	95.280	96.510	96.510	96.510	39,287.385
	21-May~20-Jun	45,020.882	45,002.670	45,002.670	45,002.670	101.420	102.650	102.650	102.650	44,900.020
	21-Jun~20-Jul	27,537.272	27,519.060	27,519.060	27,519.060	184.760	185.990	185.990	185.990	27,333.070
	21-Jul~20-Aug	37,834.912	37,816.700	37,816.700	37,816.700	102.220	103.450	103.450	103.450	37,713.250
	21-Aug~20-Sept	39,402.795	39,383.895	39,383.895	39,383.895	168.590	169.820	169.820	169.820	39,214.075
	21/-Sept~20-Oct	42,357.522	42,339.310	42,339.310	42,339.310	77.060	78.290	78.290	78.290	42,261.020
	21-Oct~20-Nov	29,227.472	29,209.260	29,209.260	29,209.260	123.590	124.820	124.820	124.820	29,084.440
	21-Nov~20-Dec	21,403.232	21,385.020	21,385.020	21,385.020	209.660	210.890	210.890	210.890	21,174.130
	21-Dec~31-Dec	12,820.100	12,814.546	12,814.546	12,814.546	44.461	44.930	44.930	44.930	12,769.616
	Subtotal	441,802.988	441,584.966	441,584.966	441,584.966	1,577.631	1,593.640	1,593.640	1,593.640	439,991.326
2014	01-Jan~20-Jan	23,309.273	23,299.174	23,299.174	23,299.174	80.838	81.690	81.690	81.690	23,217.484
	21-Jan~20-Feb	25,108.273	25,092.620	25,092.620	25,092.620	238.609	239.930	239.930	239.930	24,852.690

	21-Feb~20-Mar	34,941.123	34,925.470	34,925.470	34,925.470	90.889	92.210	92.210	92.210	34,833.260
	21-Mar~20-Apr	27,348.373	27,332.720	27,332.720	27,332.720	150.179	151.500	151.500	151.500	27,181.220
	21-Apr~20-May	46,056.643	46,040.990	46,040.990	46,040.990	136.019	137.340	137.340	137.340	45,903.650
	21-May~20-Jun	41,338.383	41,322.730	41,322.730	41,322.730	137.939	139.260	139.260	139.260	41,183.470
	21-Jun~20-Jul	24,079.703	24,064.050	24,064.050	24,064.050	195.699	197.020	197.020	197.020	23,867.030
	21-Jul~20-Aug	24,296.023	24,280.370	24,280.370	24,280.370	24.247	25.568	25.568	25.568	24,254.802
	21-Aug~20-Sept	19,020.870	19,006.890	19,006.890	19,006.890	230.759	232.080	232.080	232.080	18,774.810
	21/-Sept~20-Oct	36,410.883	36,395.230	36,395.230	36,395.230	137.779	139.100	139.100	139.100	36,256.130
	21-Oct~20-Nov	39,572.323	39,556.670	39,556.670	39,556.670	120.969	122.290	122.290	122.290	39,434.380
	21-Nov~20-Dec	54,077.303	54,061.650	54,061.650	54,061.650	41.689	43.010	43.010	43.010	54,018.640
	21-Dec~31-Dec	14,461.733	14,455.629	14,455.629	14,455.629	23.745	24.214	24.214	24.214	14,431.415
	Subtotal	410,020.906	409,834.193	409,834.193	409,834.193	1,609.360	1,625.213	1,625.213	1,625.213	408,208.980
2015	01-Jan~20-Jan	26,294.059	26,282.961	26,282.961	26,282.961	43.172	44.026	44.026	44.026	26,238.935
	21-Jan~20-Feb	23,347.362	23,330.160	23,330.160	23,330.160	128.787	130.110	130.110	130.110	23,200.050
	21-Feb~20-Mar	29,334.242	29,317.040	29,317.040	29,317.040	90.567	91.890	91.890	91.890	29,225.150
	21-Mar~20-Apr	36,445.662	36,428.460	36,428.460	36,428.460	103.347	104.670	104.670	104.670	36,323.790
	21-Apr~20-May	50,926.312	50,909.110	50,909.110	50,909.110	135.837	137.160	137.160	137.160	50,771.950
	21-May~20-Jun	44,347.872	44,330.670	44,330.670	44,330.670	93.227	94.550	94.550	94.550	44,236.120
	21-Jun~20-Jul	21,781.692	21,764.490	21,764.490	21,764.490	201.317	202.640	202.640	202.640	21,561.850
	21-Jul~20-Aug	25,669.432	25,652.230	25,652.230	25,652.230	181.047	182.370	182.370	182.370	25,469.860
	21-Aug~20-Sept	24,595.672	24,578.470	24,578.470	24,578.470	205.267	206.590	206.590	206.590	24,371.880
	21/-Sept~20-Oct	28,268.402	28,251.200	28,251.200	28,251.200	105.877	107.200	107.200	107.200	28,144.000
	21-Oct~20-Nov	32,129.982	32,112.780	32,112.780	32,112.780	188.397	189.720	189.720	189.720	31,923.060
	21-Nov~20-Dec	27,519.202	27,502.000	27,502.000	27,502.000	195.037	196.360	196.360	196.360	27,305.640

	21-Dec~31-Dec	8,920.294	8,915.599	8,915.599	8,915.599	49.378	49.894	49.894	49.894	8,865.705
	Subtotal	379,580.185	379,375.171	379,375.171	379,375.171	1,721.257	1,737.180	1,737.180	1,737.180	377,637.991
2016	01-Jan~20-Jan	16,218.716	16,210.181	16,210.181	16,210.181	89.778	90.716	90.716	90.716	16,119.465
	21-Jan~20-Feb	26,613.060	26,599.830	26,599.830	26,599.830	60.796	62.250	62.250	62.250	26,537.580
	21-Feb~20-Mar	25,531.520	25,518.290	25,518.290	25,518.290	46.666	48.120	48.120	48.120	25,470.170
	21-Mar~20-Apr	28,479.910	28,466.680	28,466.680	28,466.680	73.666	75.120	75.120	75.120	28,391.560
	21-Apr~20-May	21,821.914	21,809.560	21,809.560	21,809.560	89.386	90.840	90.840	90.840	21,718.720
	21-May~20-Jun	22,342.360	22,329.130	22,329.130	22,329.130	143.326	144.780	144.780	144.780	22,184.350
	21-Jun~20-Jul	20,878.600	20,865.370	20,865.370	20,865.370	105.836	107.290	107.290	107.290	20,758.080
	21-Jul~20-Aug	14,869.920	14,856.690	14,856.690	14,856.690	144.166	145.620	145.620	145.620	14,711.070
	21-Aug~20-Sept	13,052.190	13,038.960	13,038.960	13,038.960	180.506	181.960	181.960	181.960	12,857.000
	21/-Sept~20-Oct	16,525.900	16,512.670	16,512.670	16,512.670	97.966	99.420	99.420	99.420	16,413.250
	21-Oct~20-Nov	22,179.880	22,166.650	22,166.650	22,166.650	69.406	70.860	70.860	70.860	22,095.790
	21-Nov~20-Dec	23,288.590	23,275.360	23,275.360	23,275.360	122.376	123.830	123.830	123.830	23,151.530
	21-Dec~31-Dec	5,315.825	5,310.782	5,310.782	5,310.782	37.739	38.216	38.216	38.216	5,272.566
	Subtotal	257,118.385	256,960.153	256,960.153	256,960.153	1,261.613	1,279.022	1,279.022	1,279.022	255,681.131
2017	01-Jan~20-Jan	9,665.135	9,655.968	9,655.968	9,655.968	68.616	69.484	69.484	69.484	9,586.484
	21-Jan~20-Feb	20,822.080	20,807.870	20,807.870	20,807.870	75.165	76.510	76.510	76.510	20,731.360
	21-Feb~20-Mar	17,447.300	17,433.090	17,433.090	17,433.090	103.495	104.840	104.840	104.840	17,328.250
	21-Mar~20-Apr	23,384.030	23,369.820	23,369.820	23,369.820	99.505	100.850	100.850	100.850	23,268.970
	21-Apr~20-May	22,471.200	22,456.990	22,456.990	22,456.990	120.755	122.100	122.100	122.100	22,334.890
	21-May~20-Jun	19,204.810	19,190.600	19,190.600	19,190.600	84.495	85.840	85.840	85.840	19,104.760
	21-Jun~20-Jul	18,553.480	18,539.270	18,539.270	18,539.270	192.555	193.900	193.900	193.900	18,345.370
	21-Jul~20-Aug	16,755.110	16,740.900	16,740.900	16,740.900	200.455	201.800	201.800	201.800	16,539.100

	21-Aug~20-Sept	21,551.070	21,536.860	21,536.860	21,536.860	165.225	166.570	166.570	166.570	21,370.290
	21/-Sept~20-Oct	26,631.270	26,617.060	26,617.060	26,617.060	123.705	125.050	125.050	125.050	26,492.010
	21-Oct~20-Nov	42,839.840	42,825.630	42,825.630	42,825.630	89.045	90.390	90.390	90.390	42,735.240
	21-Nov~20-Dec	49,999.810	49,985.600	49,985.600	49,985.600	107.665	109.010	109.010	109.010	49,876.590
	21-Dec~31-Dec	9,613.209	9,605.399	9,605.399	9,605.399	41.171	41.569	41.569	41.569	9,563.829
	Subtotal	298,938.344	298,765.056	298,765.056	298,765.056	1,471.852	1,487.913	1,487.913	1,487.913	297,277.143
2018	01-Jan~20-Jan	17,478.561	17,464.361	17,464.361	17,464.361	74.856	75.581	75.581	75.581	17,388.781
	21-Jan~20-Feb	29,167.190	29,145.180	29,145.180	29,145.180	114.457	115.580	115.580	115.580	29,029.600
	21-Feb~20-Mar	21,190.440	21,168.430	21,168.430	21,168.430	95.607	96.730	96.730	96.730	21,071.700
	21-Mar~20-Apr	26,070.400	26,048.390	26,048.390	26,048.390	80.567	81.690	81.690	81.690	25,966.700
	21-Apr~20-May	24,396.580	24,376.580	24,376.580	24,376.580	52.467	53.590	53.590	53.590	24,322.990
	21-May~20-Jun	24,203.000	24,180.990	24,180.990	24,180.990	138.647	139.770	139.770	139.770	24,041.220
	21-Jun~20-Jul	21,275.670	21,253.660	21,253.660	21,253.660	325.207	326.330	326.330	326.330	20,927.330
	21-Jul~20-Aug	15,360.540	15,338.540	14,348.956	14,348.956	89.796	90.892	90.892	90.892	14,258.064
	21-Aug~29-Aug	7,255.341	7,249.541	7,249.541	7,249.541	27.372	28.208	28.208	28.208	7,221.333
	Subtotal	186,397.723	186,225.672	185,236.088	185,236.088	998.976	1,008.370	1,008.370	1,008.370	184,227.718
	Total	1,973,858.530	1,972,745.211	1,971,755.627	1,971,755.627	8,640.689	8,731.338	8,731.338	8,731.338	1,963,024.289

*Note:

1. All electricity data measured by Meter1 and Meter2 were daily recorded by project owner and grid company respectively and monthly reported. During this monitoring period, the cut-off time for electricity exports and imports was at 00:00 on the 21st day of each month, and ETNs were issued by grid company for the cross check.

2. Readings for the period 01-Jan to 20-Jan in 2013 to 2018 and 21-Dec to 31-Dec in 2013 to 2017 as well as readings for 21-29 Aug in 2018 were sourced from the daily records and verified with the confirmation letter issued by the grid company/9/. The verification team accepted this approach for ERs calculation since it is reasonable.

As stated in Section 4.2 of the MR (version 01 dated 28/04/2022), readings of electricity meters are monthly recorded, thus CL 03 was raised to clarify the source of readings for 01-Jan to 20-Jan in 2013 to 2018 and 21-Dec to 31-Dec in 2013 to 2017 as well as readings for 21-Aug to 29-Aug in 2018 and the CL has been successfully closed. Refer to Appendix D for details.

The corresponding baseline emission reductions during this monitoring period is calculated as:

Year	EG _y (MWh)	EF _y (tCO ₂ /MWh)	BE _y (tCO ₂)
2013	439,991.326	1.07460	472,814
2014	408,208.980	1.07460	438,661
2015	377,637.991	1.07460	405,809
2016	255,681.131	1.07460	274,754
2017	297,277.143	1.07460	319,454
2018 (01/01/2018 to 29/08/2018)	184,227.718	1.07460	197,971
Total	1,963,024.289	-	2,109,463

Project emissions

As statement in the registered VCS PD, for the wind power activities, the project emissions from the project is zero. Hence, PE_y during the monitoring period from Inner Mongolia Baotou Bayin Wind Power Project is zero.

Leakages

Leakage does not need to be accounted for this project as per the monitoring methodology ACM0002 (version 6.0).

Emission reductions

The emission reductions for this monitoring period were calculated as:

Monitoring period	GHG emission reductions or removals (tCO ₂ e)
01/01/2013 - 31/12/2013	472,814
01/01/2014 - 31/12/2014	438,661
01/01/2015 - 31/12/2015	405,809
01/01/2016 - 31/12/2016	274,754
01/01/2017 - 31/12/2017	319,454
01/01/2018 - 29/08/2018	197,971
Total ERs claimed (in 2,067 days)	2,109,463

Comparison of actual emission reductions or net anthropogenic GHG removals by sinks with estimates in registered CDM-PDD

The emission reductions claimed are 2,109,463 tCO₂e in this monitoring period (i.e. 2,067 days). Compared with expected emission reductions 2,835,346 tCO₂e (calculated as 500,678/365d*2,067d), the reported emission reductions in this monitoring period are 25.60% lower than the expected. Considering the natural fluctuate of the wind resource and actual generator operation conditions as well as the limitation of the grid absorption capacity, CTI is able to confirm that the actual power supply and also emission reductions reported in this monitoring period are reasonable and appropriate.

CTI verified the input data for calculating emission reductions and the calculating process, and confirmed the result were complete and transparent. Therefore, CTI is able to confirm that the actual power supply and also emission reductions reported in this monitoring period are reasonable and appropriate.

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

All necessary documentations are collected, referenced and aggregated, which is easily accessible in hard-copy or electronic format. Measurements are performed by calibrated equipment, and the key data can also be cross-checked via other sources, such as records, receipts and inventory data. No assumptions are used that have any material influence on reported emission reductions.

CTI concludes that during this monitoring period, the evidences for determination of emission reductions are sufficient and reasonable, and the calculation of emission reductions is reliable.

4.6 Non-Permanence Risk Analysis

The project is not AFOLU project, and thus non-permanence risk analysis is not applicable for the project.

5 VERIFICATION CONCLUSION

Shenzhen CTI International Certification Co., Ltd (CTI) has performed the verification of the emission reductions that have been reported for the project activity “Inner Mongolia Baotou Bayin Wind Power Project” in China (VCS Project ID: 355) for the period 01/01/2013-29/08/2018.

The verification is based on the baseline and monitoring methodology ACM0002 (version 6.0), the VCS PD. The verification consisted of the following three phases: i) desk review of the project design and the baseline and monitoring plan; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and the issuance of the final verification and certification report.

The project proponents are responsible for the collection, calculation and determination of the GHG data in accordance with the monitoring plan and the reporting of GHG emission reductions on the basis set out within the project monitoring report.

Our verification approach was based on the requirements as defined under the applicable VCS Standard Version 4.2 and relevant UNFCCC requirements. Our approach is risk-based, drawing on an understanding of the risks associated with reporting GHG emissions data and the controls in place to mitigate these. The verification can confirm that:

- the project is implemented and operated as per the registered VCS PD ;
- the monitoring plan is as per the applied methodology;
- the monitoring complies with the monitoring plan;
- the monitoring report and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS Standard Version 4.2 and CDM requirements;
- the installed equipment being essential for generating emission reduction runs reliably and is calibrated appropriately;
- the monitoring system is in place and generates GHG emission reductions data;
- the GHG emission reductions are calculated without material misstatements.

It is CTI's responsibility to provide an independent verification statement on the reported GHG emission reductions for the project. Based on an understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these, CTI planned and performed our work to obtain the information and explanations that we considered necessary to provide reasonable assurance that reported GHG emission reductions are fairly stated.

CTI does not assume any responsibility towards the issuance and utilization of the VCUs hereby verified and certified. Request for issuance of VCUs shall be made by the project proponent to an approved VCS Program Registry based on the requirements set out under the most recent version of the VCS Program Guidelines clause on VCS Registration.

The verification of reported emission reductions is based on the information made available to CTI and the engagement conditions detailed in this report. CTI cannot be held liable by any party for decisions made or not made based on this report.

In CTI's opinion the GHG emissions reductions of the "Inner Mongolia Baotou Bayin Wind Power Project" for the period 01/01/2013-29/08/2018 are fairly stated in the monitoring report (version 02 dated 30/05/2022). The GHG emission reductions were calculated correctly on the basis of the approved methodology ACM0002 (version 6.0) and the monitoring plan contained in the registered VCS PD.

CTI can confirm that the GHG emission reductions are calculated without material misstatements. Based on the evidence and information that are considered necessary to guarantee that GHG emission reductions are appropriately calculated, CTI confirms the following statement:

Reporting period: 01/01/2013-29/08/2018

Verified GHG emission reductions and removals in the above verification period:

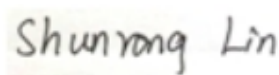
Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2013	472,814	0	0	472,814
2014	438,661	0	0	438,661
2015	405,809	0	0	405,809
2016	274,754	0	0	274,754
2017	319,454	0	0	319,454
2018				
(01/01/2018 - 29/08/2018)	197,971	0	0	197,971
Total	2,109,463	0	0	2,109,463



Mr.Feng Tian

Team Leader

01-June-2022



Ms. Lin Shunrong

Technical Reviewer

01-June-2022

APPENDIX A: ABBREVIATIONS

CAR	Corrective Action Request
CCER	Chinese Certified Emission Reduction
CER	Certified Emission Reduction(s)
CL	Clarification request
CO ₂	Carbon Dioxide
CO ₂ e	Carbon Dioxide Equivalent
CTI	Shenzhen CTI International Certification Co., Ltd
DOE	Designated Operational Entity
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reduction
ETN	Electricity Transaction Note
EU ETS	EU Emissions Trading Scheme
FAR	Forward Action Request
FSR	Feasibility Study Report
GHG	Greenhouse Gas(es)
IREC	International Renewable Energy Certificates
MR	Monitoring Report
NCPG	North China Power Grid
PD	Project Description
PP	Project Proponent
SN	Serial Number
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit

APPENDIX B: REFERENCES

Documentation used to verify the information provided by the project proponents

/1/	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.: VER/VCU Monitoring Report for Inner Mongolia Baotou Bayin Wind Power Project, version 01 dated 28/04/2022, version 02 dated 30/05/2022
/2/	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.: Emission reduction calculation spreadsheet for Inner Mongolia Baotou Bayin Wind Power Project, version 1.0 dated 28/04/2022
/3/	Longyuan (Baotou) Wind Power Co., Ltd. and Inner Mongolia Electric Power Company: Power Purchase Agreement for Inner Mongolia Baotou Bayin Wind Power Project
/4/	Longyuan (Baotou) Wind Power Co., Ltd.: Diagram of power connection system
/5/	Longyuan (Baotou) Wind Power Co., Ltd.: VER monitoring manual and management procedure.
/6/	Longyuan (Baotou) Wind Power Co., Ltd.: Records of training for on-site staff.
/7/	Longyuan (Baotou) Wind Power Co., Ltd.: Daily operation and maintenance records.
/8/	Longyuan (Baotou) Wind Power Co., Ltd.: Monthly and daily reading records of Inner Mongolia Baotou Bayin Wind Power Project from 01/01/2013-29/08/2018
/9/	Inner Mongolia Electric Power Company.: Electricity Transaction Notes from 01/01/2013-29/08/2018 & Confirmation letter for electricity readings issued by the grid company
/10/	Questionnaires filled by local stakeholders
/11/	Longyuan (Baotou) Wind Power Co., Ltd.: Registered VSC PD i.e. Registered CDM PDD (version 04 dated 25/07/2008)
/12/	Inner Mongolia Autonomous Region Quality and Technical Supervision Bureau: Accreditation certificates of Inner Mongolia Electric Power Research Institute Electric Metrological Centre, issued on 02/01/2009 and valid to 01/01/2014, 02/01/2014 and valid to 01/01/2018
/13/	Inner Mongolia Electric Power Research Institute Electric Metrological Centre: Calibration certificates for meters Meter1, Meter1', Meter2 and Meter2' covering this monitoring period
/14/	Business license of Longyuan (Baotou) Wind Power Co., Ltd.
/15/	State Economic and Trade Commission: Technical administrative code of electric energy metering (DL/T 448-2016)
/16/	UNFCCC, http://cdm.unfccc.int

/17/	Gold Standard, http://www.goldstandard.org/
/18/	China CCER Exchange Info-platform, http://cdm.ccchina.org.cn/ccer.aspx
/19/	Measures for the Administration of Carbon Emission Trading, issued by Ministry of Ecology and Environment of China, dated 31/12/2020
/20/	Enforced company list in China cap & trade scheme, issued by Ministry of Ecology and Environment of China
/21/	Longyuan (Baotou) Wind Power Co., Ltd.: Statement on the annual income of employees from 2013 to 2018.
/22/	Per capita disposable income of residents in Inner Mongolia Autonomous Region in 2013 and 2018 https://www.nmg.gov.cn/
/23/	Bureau Veritas Certification Holding SAS: VCS Validation Report
/24/	DET NORSKE VERITAS: CDM Validation Report, version 05, dated 12/05/2009
/25/	VER/VCU Monitoring Report for Inner Mongolia Baotou Bayin Wind Power Project of previous monitoring period 30/08/2008-11/05/2009
/26/	Bureau Veritas Certification Holding SAS: VCS Verification Report for Inner Mongolia Baotou Bayin Wind Power Project of previous monitoring period 30/08/2008-11/05/2009
/27/	CDM/CER Monitoring Reports for Inner Mongolia Baotou Bayin Wind Power Project of previous CDM monitoring periods 12/05/2009-30/06/2009, 01/07/2009-31/12/2009, 01/01/2010-30/04/2010, 01/05/2010-30/11/2010, 01/12/2010-30/04/2011, 01/05/2011-30/04/2012 and 01/05/2012-31/12/2012
/28/	CDM Verification Reports for Inner Mongolia Baotou Bayin Wind Power Project of previous monitoring period 12/05/2009-30/06/2009, 01/07/2009-31/12/2009, 01/01/2010-30/04/2010, 01/05/2010-30/11/2010, 01/12/2010-30/04/2011, 01/05/2011-30/04/2012 and 01/05/2012-31/12/2012
/29/	Xinjiang Gold Wind Science and Technology Co., Ltd.: Nameplates of the equipment
/30/	Feasibility Study Report of Inner Mongolia Baotou Bayin Wind Power Project
/31/	Inner Mongolia Electric Power Company: Statement on the deactivation of the 10kV backup line
/32/	Longyuan (Baotou) Wind Power Co., Ltd.: Statement on the timeline of the Project
/33/	Co-signed by the Xinjiang Gold Wind Science and Technology Co., Ltd. and Longyuan (Baotou) Wind Power Co., Ltd.: Wind Turbines Purchase Agreement, dated 10/04/2007
/34/	Longyuan (Baotou) Wind Power Co., Ltd.: Statement on no double counting

/35/	Law of the People's Republic of China on Promotion of Employment Order of the President of the People's Republic of China
/36/	Action Plan for Carbon Dioxide Peaking Before 2030 published by the State Council on 24/10/2021
/37/	Responding to Climate Change: China's Policies and Actions released by the State Council on 10/2021

Methodologies, tools and other guidance

/38/	Verified Carbon Standard: VCS Standard, version 4.2
/39/	Verified Carbon Standard: VCS Program Guide, version 4.1
/40/	Verified Carbon Standard: VCS Sectoral Scopes http://v-c-s.org/node/448
/41/	Verified Carbon Standard: Registration and Issuance Process, version 4.1
/42/	UNFCCC EB: Approved methodology, ACM0002 , version 6.0

Persons interviewed

	Interviewed personnel	Role	Organization
/43/	Mr. Zhang Bin	Plant Director	Longyuan (Baotou) Wind Power Co., Ltd.
	Mr. Li lei	O&M Staff	
	Mr. Li Xiaolei	O&M Staff	
	Mr. Chen Xijin	Local resident	Local Village
	Mr. Liu Jianmin		
	Mr. Huang Puyi	Staff	Local Environment Protection Bureau
	Mr. Xue Qiang	Project Manager	Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.

APPENDIX C: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Mr. Tian FENG

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification					
GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
√	-	√	√	-	-

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu LIN

Wu Lin

Technical Competent Manager

Shenzhen, 10/02/2022

Ms. Shunrong LIN

Satisfies the requirements of competence management system of CTI Certification, and is hereby appointed as:

Qualification						
Status	GHG Auditor	Validator	Verifier	Team Leader	Technical Reviewer	Technical Expert
Date	√	√	√	√	√	√

Scope	Technical Area
SS 1: Energy industries (renewable/non-renewable sources)	TA 1.2: Energy generation from renewable energy sources
SS 14: Afforestation and reforestation	TA 14.1: Afforestation and reforestation
SS 15: Agriculture	TA 15.1: Agriculture

This appointment is valid for 3 years from its date of approval below and is bound by internal requirements of management system of the Certification Body of CTI.

Approved by:

Wu Lin

Wu LIN

Technical Competent Manager

Shenzhen, 01/01/2021

APPENDIX D: CORRECTIVE ACTION REQUESTS, CLARIFICATION REQUESTS AND FORWARD ACTION REQUESTS

Table 1: Corrective Action Requests

CAR ID	Corrective Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA

Table 2: Clarification Requests

CL ID	Clarification Request	Response by Project Proponent	Verification Team Assessment
CL 01	The starting date of operation for the first wind turbine stated in MR (version 01 dated 28/04/2022) is inconsistent with the previous CDM verifications, thus, CL 01 was raised for clarification	The monitoring reports of previous CDM verifications stated that the project the first wind turbine started operation on 31-August-2008. However, according to the Statement on the project timeline, the first wind turbine started trail operation on 30-August-2008 and was put into official operation on 31-August-2008. Thus, start date of operation for the first wind turbine is confirmed to be 30-August-2008 on which the project starting trail operation.	OK. It has been revised and verified by CTI by checking updated MR. Therefore, CL 01 is closed.

CL ID	Clarification Request	Response by Project Proponent	Verification Team Assessment
CL 02	Type of electricity meters is missing in MR (version 01 dated 28/04/2022) thus, CL 02 was raised for clarification	The type of meters had been included in the updated in the MR.	OK. It has been revised and verified by CTI by checking updated MR. Therefore, CL 02 is closed.

CL ID	Clarification Request	Response by Project Proponent	Verification Team Assessment
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CL 03	As stated in Section 4.2 of the MR (version 01 dated 28/04/2022), readings of electricity meters are monthly recorded, thus CL 03 was raised to clarify the source of readings for 01-Jan to 20-Jan in 2013 to 2018 and 21-Dec to 31-Dec in 2013 to 2017 as well as readings for 21-Aug to 29-Aug in 2018	Readings for 01-Jan to 20-Jan in 2013 to 2018 and 21-Dec to 31-Dec in 2013 to 2017 as well as readings for 21-Aug to 29-Aug in 2018 are sourced from daily reading records and have been confirmed by grid company.	OK. It has been revised and verified by CTI by checking updated MR and Electricity confirmation letter issued by grid company. Therefore, CL 03 is closed.
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Table 3: Forward Action Requests

FAR ID	Forward Action Request	Response by Project Proponent	Verification Team Assessment
NA	NA	NA	NA